Option Portfolio Selection Around Two Event Shocks

SPY Tariffs, USO Conflict Risk, and Volatility Forecasts

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Abstract

This paper studies option portfolios entered before two event shocks: the April 2025 tariff escalation in SPY and the February–March 2026 US–Iran conflict in USO. It asks two trading questions. Given probabilities for an event and an opposing outcome, which bounded option portfolio offers the preferred directional exposure? Given no directional view but a forecast of realized volatility, which daily delta-hedged gamma position is preferred, and are the entry options rich or cheap relative to that forecast? Each study begins with ten liquid listed American options. A portfolio compiler enumerates one-to-four-leg positions, applies account limits, and sends the retained candidates through pathwise American valuation, exercise, assignment, settlement, hedging, financing, and transaction costs. Historical prices are used for the entry state and one subsequent replay; candidate selection uses only the forecasts and generated paths. The directional tariff forecast selects concentrated downside convexity, while the conflict forecast selects a capped call spread at its stated probability. The volatility analysis changes sign as the target moves through the entry implied-volatility level. Deep Sets and a typed graph network are then tested as supervised surrogates for the path evaluator. Both interpolate well within one decision date, but neither recovers a held-out optimum in the chronological development test. Direct compiled evaluation is also faster at the portfolio counts studied here. The empirical results support the portfolio construction and valuation procedure. Trading-performance and learned-search claims require more independent decision dates.

1 The study

The first decision is dated 1 April 2025. SPY closes at 561.03, the selected 11 April option expiry has a median implied volatility of 22.20%, and the tariff announcement is due the next day. The second decision is dated 27 February 2026. USO closes at 81.93, the selected 20 March expiry has a median implied volatility of 61.83%, and US–Israeli strikes begin the following morning.

For each decision, the study produces two menus. The directional menu combines a baseline outcome and an event outcome at an analyst-supplied probability. The volatility menu fixes the median horizon return at zero, varies annualized realized volatility, and applies a daily delta hedge. Cash is available in both menus with utility zero.

The recorded event path is kept out of portfolio selection. It answers the narrow question "what would this marked position have done over these dates?" One replay cannot estimate expected return, a Sharpe ratio, or performance across market regimes.

The analyst makes one discretionary choice at the entry close. Later points in the rollout are the prescribed hedge, exercise, assignment, and close actions. Re-selecting the portfolio through time would require a dated series of forecast probabilities; no such series is assumed here.

2 What happened

2.1 Tariffs: 1–9 April 2025

The 2 April order set a 10% additional tariff on most imports from 5 April and higher country-specific rates from 9 April. China announced a 34% retaliatory tariff on 4 April. The United States raised its additional China rate from 34% to 84% on 8 April. On 9 April the country-specific rates for most trading partners were paused for 90 days, the 10% general tariff remained, and the China rate rose again to 125% [4]. The S&P 500 gained 9.5% on the pause day but remained below its 2 April close [5].

The equity response was broad. All eleven sector SPDRs lost value over the 2–9 April window; XLE fell 12.5%, XLB 7.2%, XLRE 6.8%, and XLK 2.7%. The cross-sectional mean of the 10-day price-momentum score fell from −0.42 on 3 April to −5.24 on 9 April. SPY fell from the entry close of 561.03 to 496.74 before rebounding to 543.37.

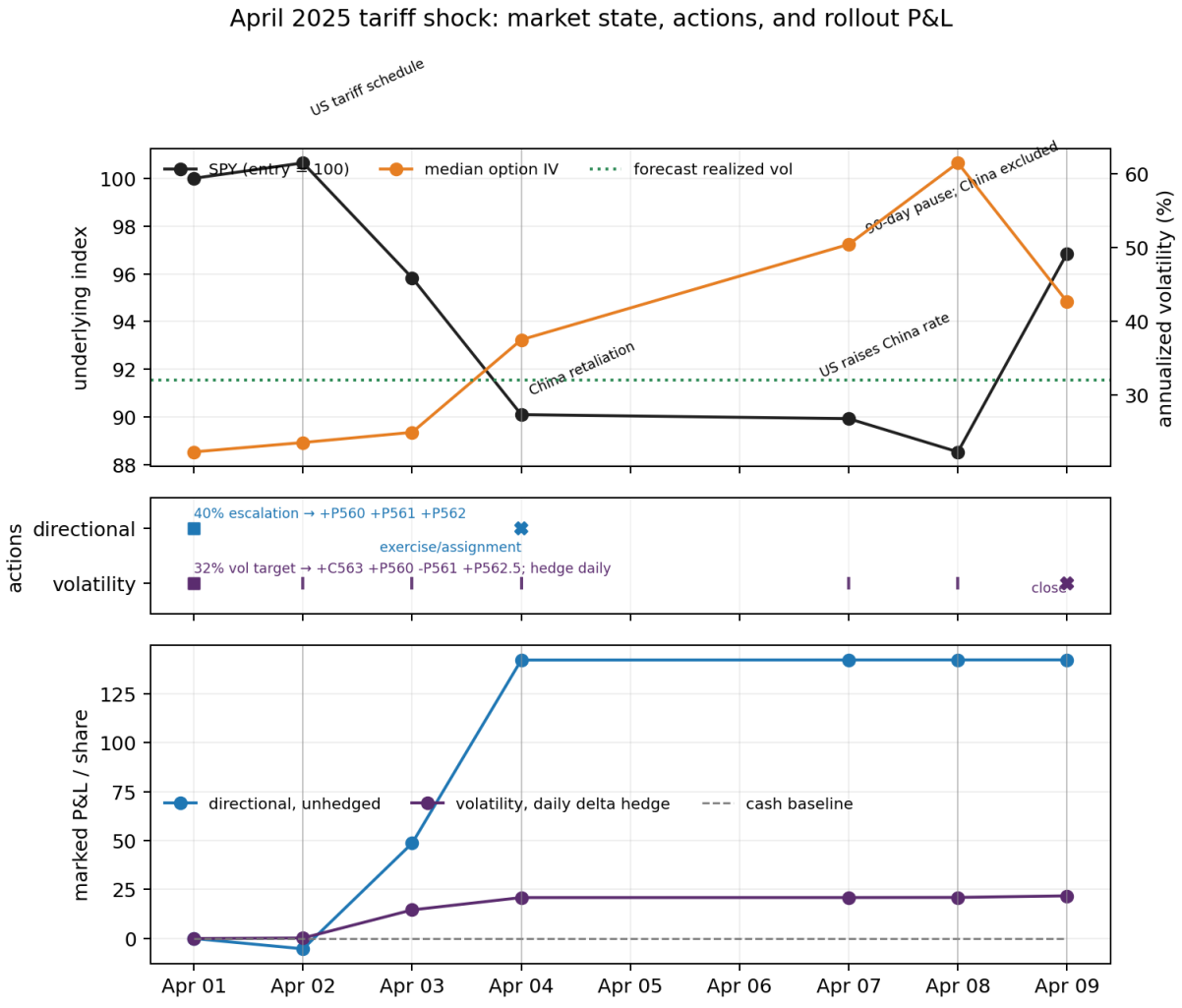


Figure 1: Tariff chronology and full marked rollout. The upper panel combines the indexed SPY close with median implied volatility. The middle panel records entry, scheduled hedges, and exit. The lower panel compares cash with the selected unhedged directional portfolio and the selected daily delta-hedged volatility portfolio.

The common date axis makes the market mechanics visible. By 4 April SPY is about ten percent below entry and median IV has risen from 22.20% to roughly 39%. The three deep-in-the-money puts exercise at that point, locking the directional result before the 9 April rebound. The delta-hedged volatility position earns about 21 per

share during the same spot and IV shock and then changes little. Cash remains at zero. A terminal payoff chart alone would miss the exercise date, the IV expansion, and the different management rules.

2.2 Conflict: 27 February–13 March 2026

Joint US–Israeli strikes began on Saturday, 28 February. Iran responded against Israel and US facilities across the Gulf and asserted control over the Strait of Hormuz. The first listed-market response came on Monday, 2 March. Oil moved above 100 dollars per barrel on 9 March amid estimates of roughly seven million barrels per day of regional production at risk [6, 7].

The market response was concentrated rather than uniform. USO rose from 81.93 on 27 February to 119.84 on 13 March. The cross-sectional dispersion of the equity volatility score rose 27% through 9 March, while its mean declined. The 10-day price-momentum mean initially rose from 0.33 to 0.79, turned negative on 4 March, and reached −2.13 by 13 March. Those factor statistics describe the broader equity distribution; the option decision below concerns USO.

These dates organize descriptive market replays. Estimating the causal effect of either event would require controls for concurrent news and market conditions.

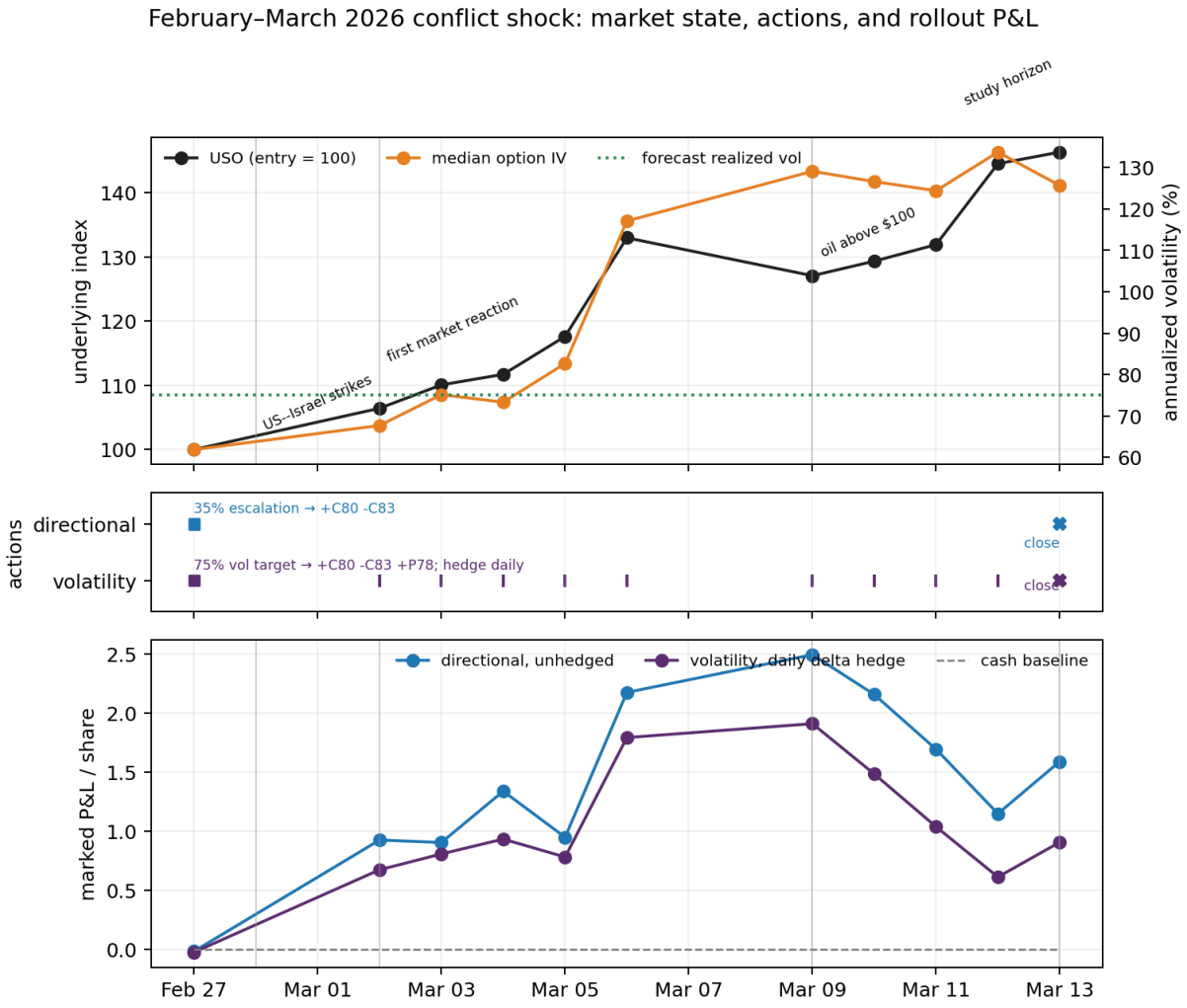


Figure 2: Conflict chronology and full marked rollout. Market state, portfolio actions, cash, the selected directional position, and the selected volatility position share one date axis.

Figure 2 shows why a correct directional call can still have a non-monotone marked result. The 80/83 call spread reaches about 2.49 per share on 9 March, then falls to 1.59 even as USO rises further. Once both calls are deep in the money, the spread converges toward its three-point cap while time and the relative marks of its two legs change. The delta-hedged volatility position follows the same early shock, peaks near 1.91, and closes at 0.91. The common timeline separates a correct market direction from the path of a particular option structure.

3 Market data and option valuation

The underlying close and option state are synchronized at 16:00 New York. The chain filter selects five calls and five puts from one expiry using only entry-date moneyness, volume, and transaction count. Every contract in the study is a physically settled American option with multiplier 100.

Some option series do not print at exactly 16:00. For each contract and day, the loader takes the implied volatility from the last valid regular-session minute and revalues the contract at the synchronized underlying close. A same-right strike interpolation supplies the volatility when the selected contract has no valid minute observation. The recorded option marks and their source files are fingerprinted.

American contracts are valued with a Cox–Ross–Rubinstein lattice [1]. At every path observation the evaluator checks the exercise value against continuation value. Long exercise and short assignment alter cash and underlying inventory; physical settlement and the fixed-horizon close are then applied. A European contract is the special case whose exercise set contains only expiry. The present empirical chains contain American contracts throughout.

The option-cost assumption is 0.65 dollars per contract side. The delta-hedged study also charges one basis point for underlying trades. Results are reported in option-price units per share; multiplying by the contract multiplier converts one unit of quantity to dollars. The marks do not reconstruct a fill at the bid or offer, so the reported replay is a marked implementation study rather than an executable fill record.

	Tariff study	Conflict study
Underlying	SPY	USO
Entry close	1 April 2025	27 February 2026
Evaluation horizon	9 April 2025	13 March 2026
Option expiry	11 April 2025	20 March 2026
Entry spot	561.03	81.93
Median entry IV	22.20%	61.83%
Contracts	five calls, five puts	five calls, five puts
Raw candidates	2,890	2,890
Directional candidates	1,675	598
Volatility candidates	789	414

Table 1: Point-in-time decision states. Different retained counts reflect the same account limits applied to different marks and payoff geometry.

4 Portfolio construction and objectives

4.1 From contracts to candidates

Let $q_j \in \{-1, 0, +1\}$ be the quantity of contract j. A candidate is the nonzero vector $q = (q_1, \ldots, q_J)$ with at most four active entries. Canonicalization means that quantities are keyed by contract identifier, duplicate references to one contract are netted, zero positions are removed, and the remaining legs are sorted. Two input lists that differ only in leg order therefore produce one candidate.

The compiler rejects unbounded expiration loss and expiration loss above five percent of entry spot. The volatility

menu adds

$$|\Delta_0(q)| \le 0.20, \qquad |\Gamma_0(q)| \ge 10^{-5}, \qquad |\mathcal{V}_0(q)| \ge 10^{-4},$$

where vega is measured per one-point change in absolute volatility. Delta, gamma, theta per calendar day, and vega are signed sums of the contract sensitivities. Strategy names are descriptive labels applied after selection; they do not restrict the search. Expiration diagrams use the standard long/short call and put payoff conventions [8].

4.2 Directional forecast

The two directional outcomes in Table 2 specify a median horizon return, annualized spot volatility, and terminal multiplier for the entry implied-volatility surface. Each outcome contributes 256 paths. Spot follows geometric Brownian motion at the observed dates. A common, mean-reverting implied-volatility factor has annual volatility 1.0, mean reversion 8.0, and correlation -0.60 with the spot shock. The portfolio is held without a delta hedge, subject to the American lifecycle rules.

	Baseline outcome	Event outcome
Tariff	60% probability; +1% median SPY return; 16% volatility; terminal IV multiplier 0.90	40% probability; −7% median SPY return; 32% volatility; terminal IV multiplier 1.30
Conflict	65% probability; −5% median USO return; 45% volatility; terminal IV multiplier 0.85	35% probability; +20% median USO return; 75% volatility; terminal IV multiplier 1.25

Table 2: Declared forecast inputs. They are decision assumptions, not parameters estimated from the subsequent event path.

For candidate q, pathwise net P&L at the fixed horizon is $P(q, \omega)$. The ranking statistic under the stated outcome probabilities is

$$U(q) = \mathbb{E}[P(q,\omega)] + 0.25\ \mathrm{ES}_{0.10}[P(q,\omega)],$$

where $\mathrm{ES}_{0.10}$ is the signed mean of the lowest ten percent of P&L. Cash has $U = 0$. The 0.25 coefficient is a declared risk preference.

4.3 Direction-neutral volatility forecast

The volatility menu uses the same entry chain and candidate compiler, sets the median horizon return to zero, and holds each contract's entry implied volatility constant along a path. Annualized realized volatility is scanned from 0.45 to 1.60 times the median entry IV, with the event-study target inserted into the grid. Each target has 64 generated paths. The option position is delta-hedged at every daily observation. The target levels share the same standardized spot shocks, so movement across the scan comes from the volatility assumption rather than a new Monte Carlo sample.

This setup gives "rich" and "cheap" precise meanings. A target below the entry IV treats the options as rich relative to the forecast; a target above entry IV treats them as cheap. The conclusion is conditional on the chosen target, constant-IV marking rule, hedge frequency, transaction costs, and candidate limits. Gamma P&L, theta P&L, vega P&L, the delta hedge, financing, costs, and the lattice residual reconcile to net P&L on every path. "Rich" and "cheap" are shorthand for the target's relation to median IV; every leg is still valued from its own entry IV and strike.

5 Surrogate models

The path evaluator supplies four labels for each portfolio:

$$y(q) = \left(\mathbb{E}[P],\ \mathrm{ES}_{0.10}[P],\ \mathrm{sd}(P),\ \Pr(P > 0)\right).$$

The learned models estimate these labels. Utility is calculated from the first two predicted components; it is not a separate training target.

For Deep Sets [2], each active option leg has a 14-element feature vector containing signed quantity, call/put, log moneyness, mark divided by spot, implied volatility, liquidity, multiplier, exercise and settlement flags, and the leg's signed entry Greeks. Shared context contains time to expiry, rate, dividend yield, and the forecast fields. The network is

$$\widehat{y}(q, c) = \rho\left(\sum_{j:q_j \neq 0} \phi(x_j), c\right),$$

where ϕ and ρ are two-layer SiLU multilayer perceptrons with 48 hidden units. The sum makes the prediction invariant to leg order.

The relation graph contains a portfolio node, an underlying node, an expiry node, and one option-position node per active leg. Directed typed edges represent *position in portfolio*, *written on underlying*, and *expires at*, together with their reverse directions. Position nodes carry the same market and Greek information as the set model, while portfolio, underlying, and expiry data reside on their own nodes. Two message-passing layers use relation-specific weights [3]:

$$m_v^{(k)} = \sum_{(u,v,r)\in E} W_r^{(k)} h_u^{(k)}, \qquad h_v^{(k+1)} = \mathrm{LN}\Big(h_v^{(k)} + \mathrm{SiLU}(W_0^{(k)} h_v^{(k)} + m_v^{(k)})\Big).$$

The four outputs are read from the final portfolio-node state and the shared context. Hidden width is 20. This graph can later represent multiple underlyings, expiries, inventory, exercise opportunities, and account relations without flattening them into one leg set. In the present single-underlying, single-expiry portfolios, Deep Sets already matches the mathematical symmetry of the problem.

The candidate-holdout experiment initially evaluates 20% of the portfolios, uses part of that set for early stopping, and asks the model to select another 10% for path evaluation. Smooth-L1 loss fits standardized labels with AdamW. A ridge regression over summed leg features supplies a linear baseline. The chronological experiment trains on one earlier decision date, selects the checkpoint on a middle date, and evaluates the final date once.

The compiler, path engine, and learned ranking have separate jobs:

$$\text{contracts and limits} \longrightarrow \text{candidate portfolios} \longrightarrow \text{pathwise labels} \longrightarrow \text{learned shortlist}.$$

There is no reinforcement-learning result in this paper. The models here are supervised regressors over a fixed action set.

6 Tariff results

6.1 Directional view

At 40% escalation probability, exhaustive evaluation and all three surrogates select

$$+P(560) + P(561) + P(562).$$

The entry debit and expiration maximum loss are 24.1600 per share. Entry delta is -1.4571, gamma 0.0660, daily theta -1.2099, and vega 1.1069 per volatility point. Scenario-mixture utility is 14.3761.

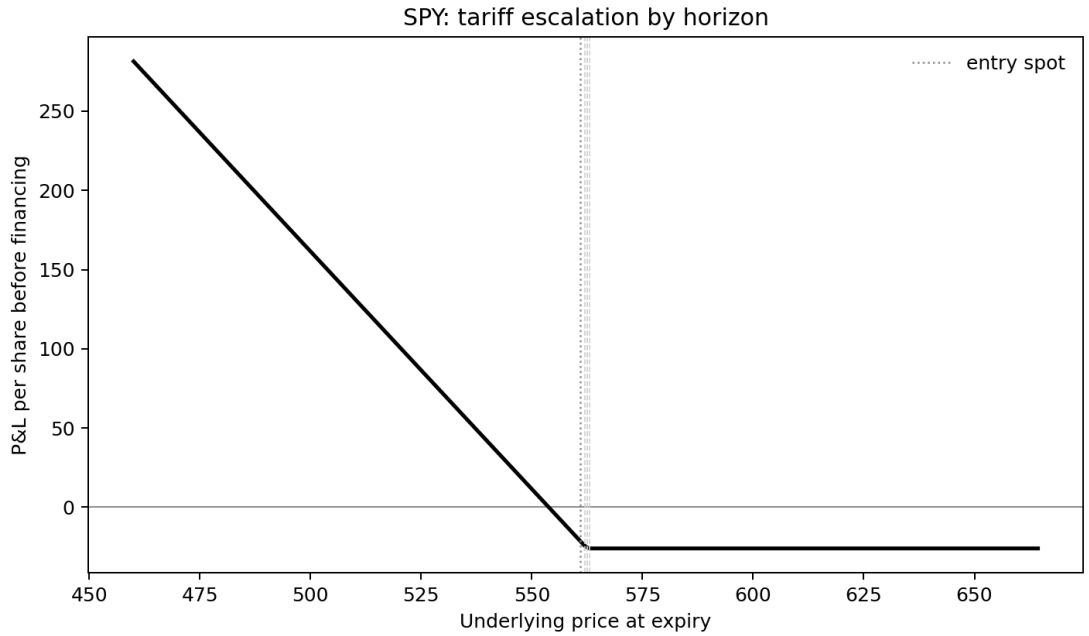


Figure 3: Expiration P&L of the tariff-escalation portfolio. Three adjacent long puts concentrate downside convexity and cap loss at the premium paid.

Under the pure escalation outcome, the preferred strikes move to 562, 562.5, and 563. Conditional mean P&L is 69.4431, the lowest-decile mean is −15.9433, and utility is 65.4573. Under the baseline outcome the same position has mean P&L of −12.2214. It first beats cash on the one-percentage-point probability grid at 23% escalation probability.

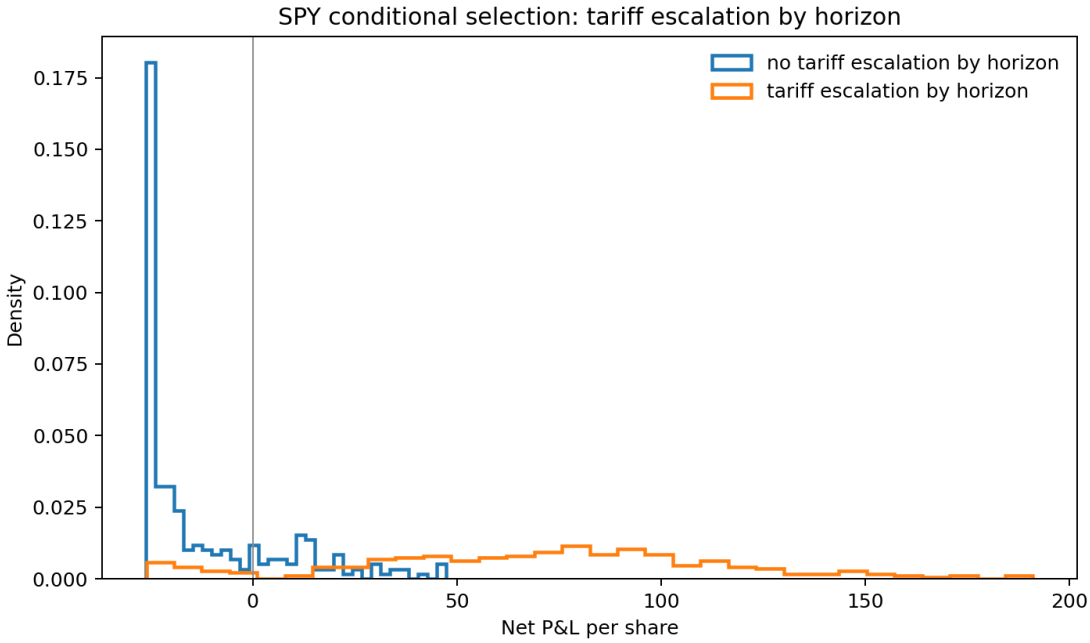


Figure 4: Fixed-horizon P&L of the pure-escalation tariff portfolio under the opposing forecast outcomes.

The 40%-mixture position earns 142.3710 per share on the recorded path. All three puts exercise during the selloff, so the lifecycle ledger does not treat them as live options only at the 9 April rebound. The pure-escalation portfolio earns 145.0551 and is also the hindsight leader. Forecast utility and recorded-path P&L have Spearman correlation 0.9478 across the retained portfolios.

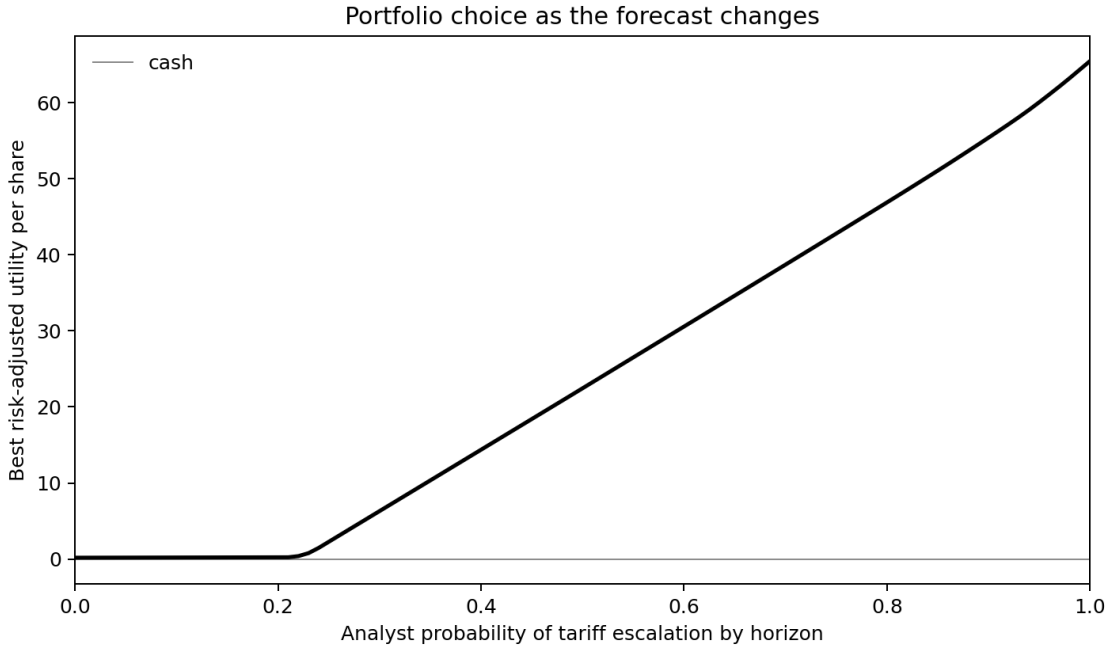


Figure 5: Best directional utility as tariff-escalation probability changes. Breakpoints mark changes in the preferred listed portfolio.

6.2 Volatility without a directional view

At entry, the median IV is 22.20%. Close-to-close realized volatility over the subsequent observations is 88.70%. The explicit 32% target from the tariff event state is above the entry IV, so this forecast treats convexity as cheap. The preferred delta-hedged position is

$$+C(563) + P(560) - P(561) + P(562.5).$$

Its entry delta is -0.0256, gamma is 0.01929, daily theta is -0.8242, and vega is 0.7371. Forecast utility is 2.5318, and recorded net P&L is 21.7708 per share. On the recorded path, gamma contributes 34.4177, vega 6.0007, theta -2.6304, and the delta hedge -19.8633, before the remaining financing, residual, and cost components.

The unrestricted optimum is short gamma through the 25.53% grid point and long gamma from 28.87% upward. The switch occurs above median entry IV because skew, bounded-loss geometry, costs, the lower-tail penalty, and daily hedge error enter alongside the scalar volatility comparison.

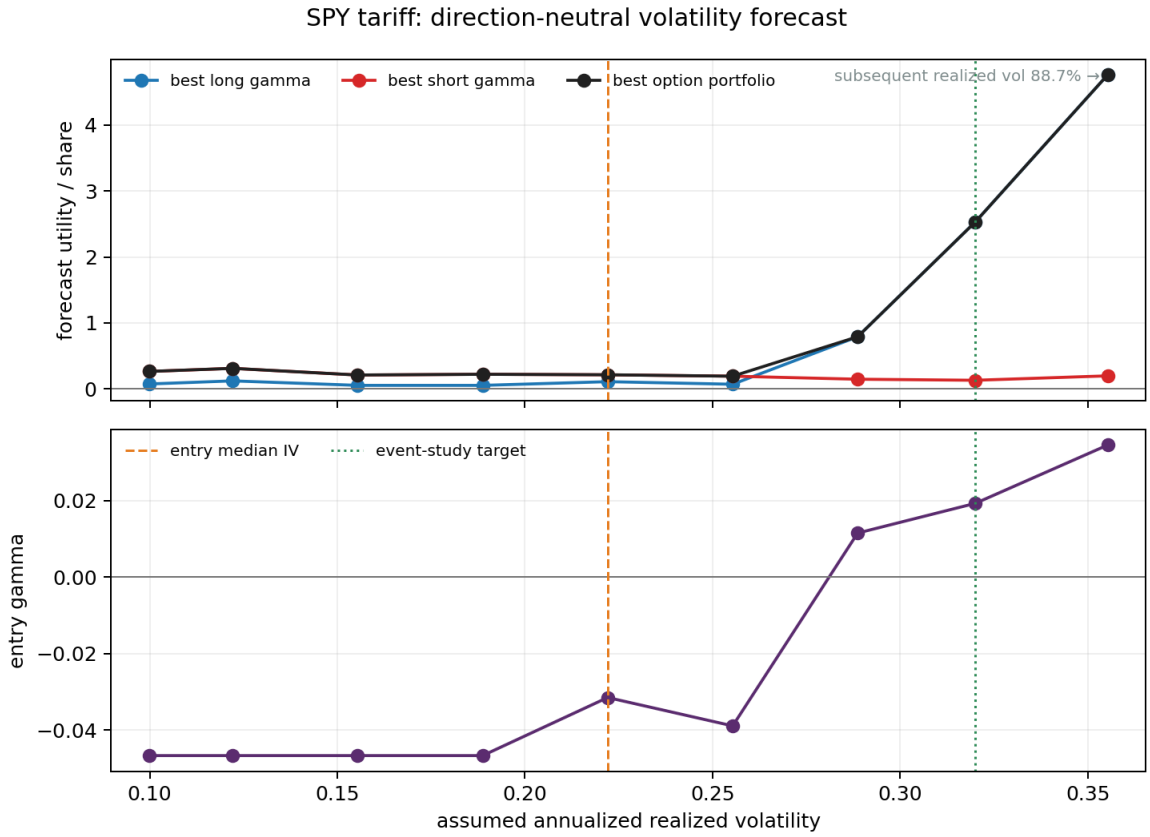


Figure 6: SPY target-volatility scan. The upper panel compares the best long-gamma, short-gamma, and unrestricted option candidates with cash. The lower panel shows the entry gamma of the selected candidate.

7 Conflict results

7.1 Directional view

At 35% escalation probability, the preferred portfolio is

$$+C(80) - C(83).$$

The debit and expiration maximum loss are 1.0600 per share; maximum expiration profit is 1.9400. Forecast utility is 0.1318, and the recorded-path P&L is 1.5902.

The pure escalation outcome selects

$$+C(80) - C(83) + P(76) - P(80).$$

The call vertical supplies capped upside. The short 80 put and long 76 put contribute premium and bounded downside. Entry delta is 0.2274, gamma 0.000073, daily theta 0.0332, and vega -0.0126 per point. This is primarily a directional structure rather than a long-volatility position.

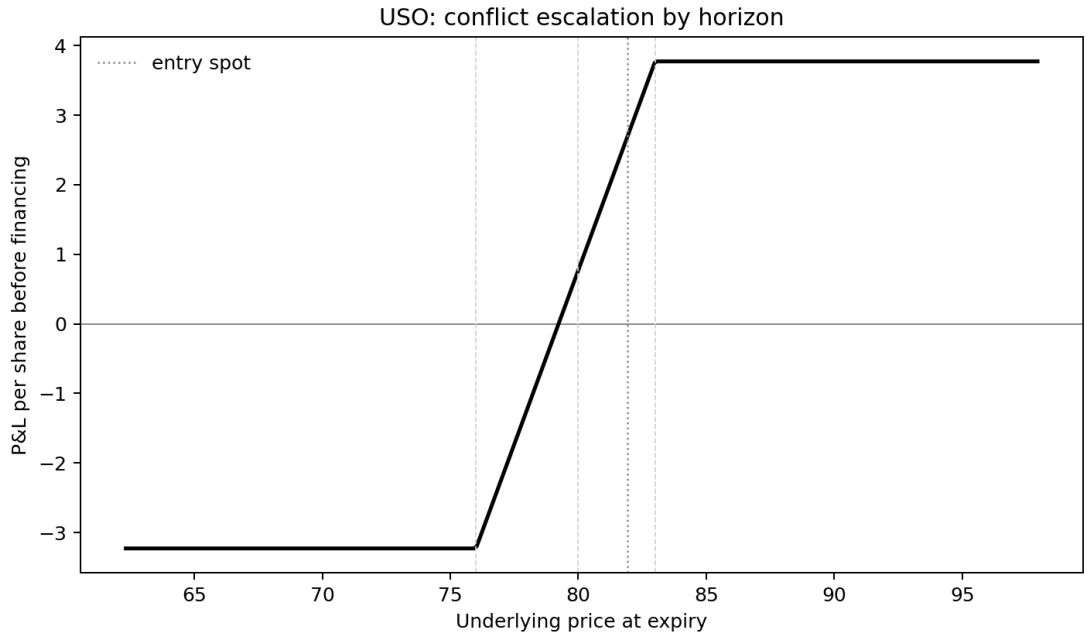


Figure 7: Expiration P&L of the pure-escalation conflict portfolio. Two bounded vertical components form the payoff between strikes 76 and 83.

Conditional mean P&L is 2.7950 under escalation and -0.2526 under the baseline. Utility is 2.6511, maximum loss is 3.2280, and the position's risk-adjusted break-even escalation probability is 34%. Other portfolios lead between 34% and 41%; the four-leg position becomes the global choice at 42%. Cash leads through 17%.

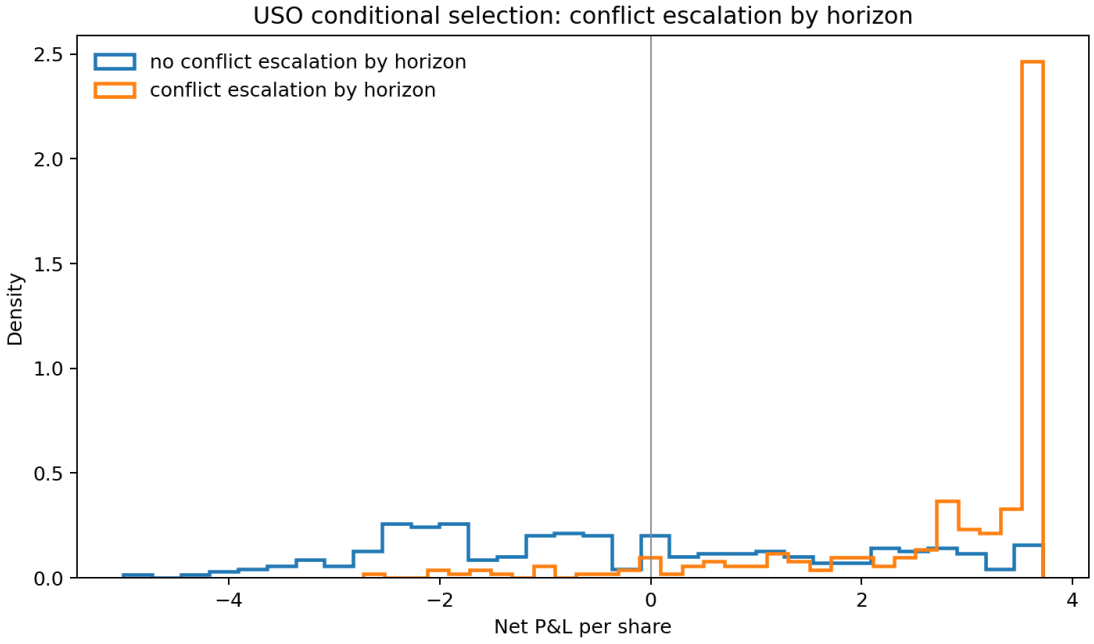


Figure 8: Fixed-horizon P&L of the four-leg conflict portfolio under the opposing forecast outcomes.

The four-leg position earns 3.3640 per share on the recorded path and is the hindsight leader. Forecast utility and

recorded P&L have Spearman correlation 0.9375 across the retained universe. The stated 35% policy chooses the cheaper call spread; the larger four-leg position enters the menu only as escalation confidence rises.

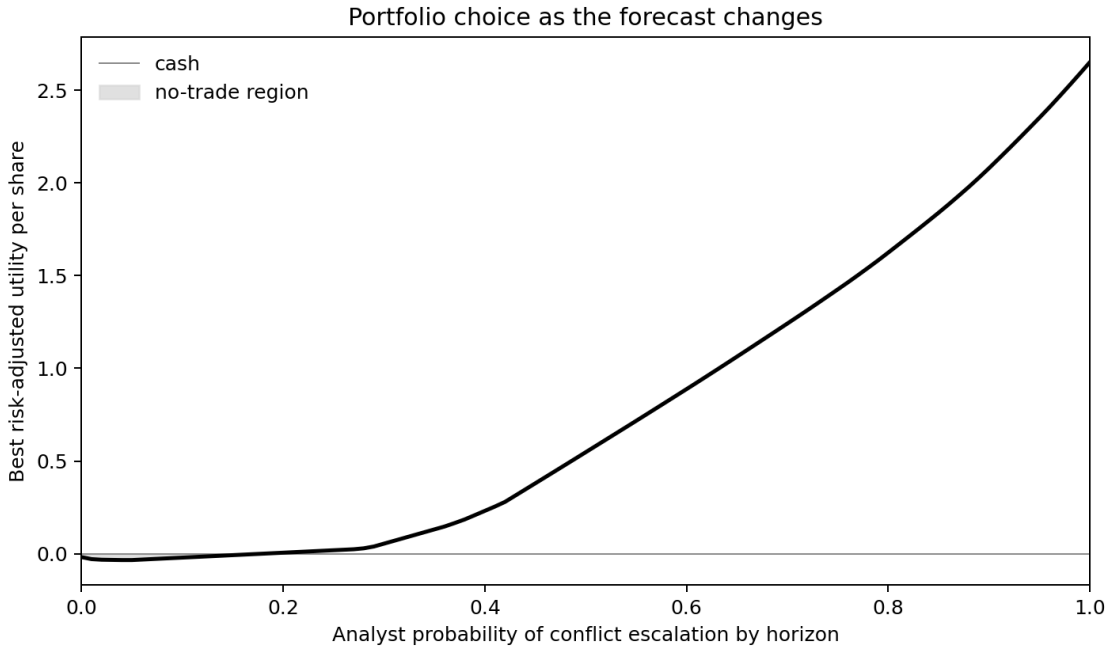


Figure 9: Best directional utility as conflict-escalation probability changes. Cash is the policy at low probabilities.

7.2 Volatility without a directional view

USO enters at 61.83% median IV. Subsequent close-to-close realized volatility is 75.48%. A 75% target is above entry IV and therefore treats the option surface as cheap relative to the forecast. The selected delta-hedged position is

$$+C(80) - C(83) + P(78).$$

Entry delta is -0.1841, gamma 0.08543, daily theta -0.08525, and vega 0.06673. Forecast utility is 0.6812, and recorded net P&L is 0.9082 per share. Recorded gamma P&L is 6.4041, theta is -1.2409, vega is 1.4849, and the delta hedge contributes 2.1560 before the other reconciling components.

The low-volatility comparison is economically different. At a target of 34.01%, or 55% of entry IV, the options are rich relative to the forecast. The preferred structure is $+C(80) - C(82) + P(76) - P(80)$, with entry gamma -0.2529. It has forecast utility 0.3615, but loses 1.5263 on the recorded path as realized volatility reaches 75.48%. Its observed gamma attribution is -9.5248, partly offset by 0.7749 of theta. This is the practical meaning of conditional valuation: the short-gamma trade is attractive under the low-vol forecast and poor under the volatility that followed.

The unrestricted optimum is short gamma at the 61.83% entry-IV grid point and long gamma at 71.11%. The 75% forecast and subsequent 75.48% realized volatility sit on the long-gamma side of that switch.

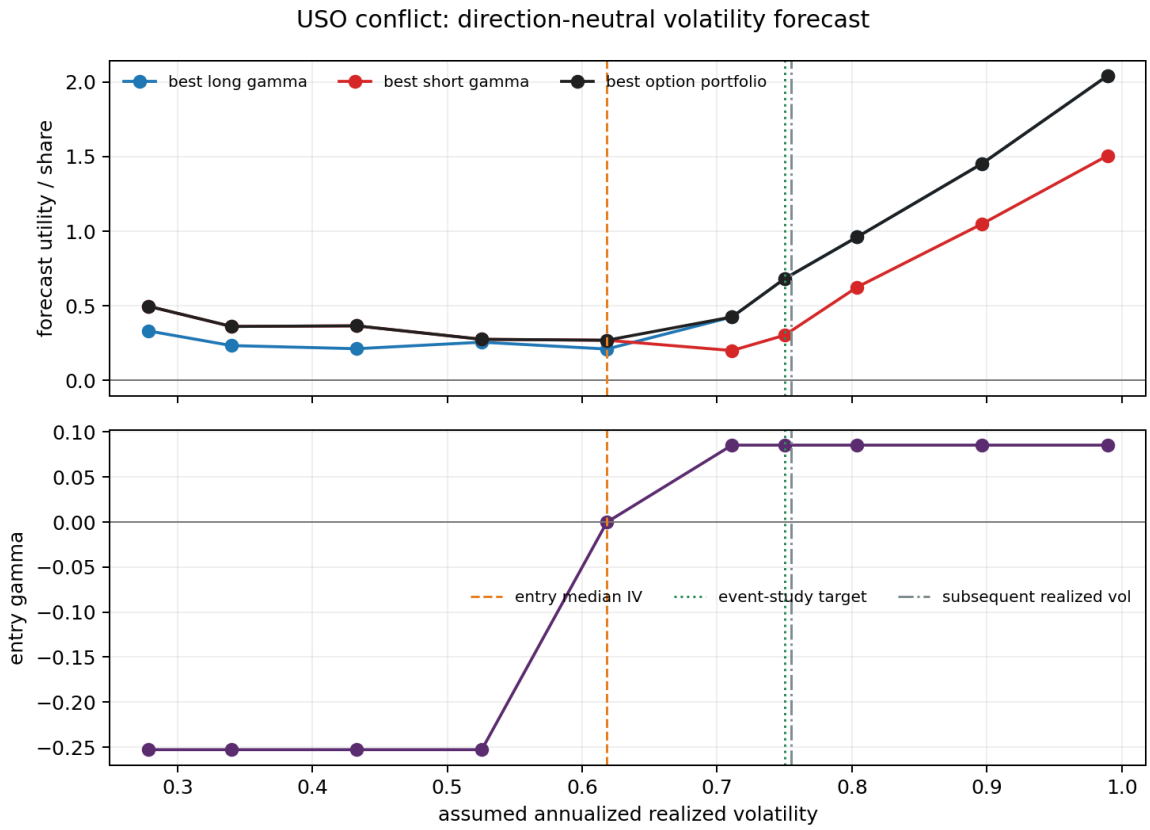


Figure 10: USO target-volatility scan. The preferred exposure changes as the assumed realized volatility moves through the entry IV.

8 Surrogate tests

8.1 Candidates held out within one date

Study	Model	Parameters	Spearman	Top-20	Fit seconds
Tariff	Deep Sets	8,596	0.9898	20/20	0.836
Tariff	Relation GNN	7,644	0.9843	20/20	13.929
Tariff	Ridge sums	112	0.9685	20/20	0.003
Conflict	Deep Sets	8,596	0.9715	20/20	0.417
Conflict	Relation GNN	7,644	0.9213	15/20	1.025
Conflict	Ridge sums	112	0.8232	14/20	0.003

Table 3: Single-seed candidate holdout under a common 30% label budget: 20% initially evaluated and 10% acquired by predicted rank.

All three models recover the tariff optimum after acquisition. The conflict optimum happens to be present in the initial 20%, so zero regret there does not test acquisition. These splits show interpolation among portfolios that share one chain, spot, horizon, and scenario definition.

8.2 A later decision date

The chronological test changes the market state. For tariffs, 1 April trains the models, 7 April selects the checkpoint, and 8 April is the final evaluation. For conflict, the corresponding dates are 27 February, 10 March, and 12 March. Each date selects its liquid contracts and recompiles its candidate universe.

Study	Model	Spearman	Top-20	Exact best	Mean regret	Fit sec.
Tariff	Deep Sets	0.5166	0.0000	0/3	5.7735	1.154
Tariff	Relation GNN	0.7486	0.1000	0/3	0.5222	31.586
Tariff	Ridge sums	−0.0874	0.0333	0/3	5.2779	0.002
Conflict	Deep Sets	−0.1053	0.0167	0/3	0.3304	0.350
Conflict	Relation GNN	0.5476	0.1667	0/3	0.3240	14.013
Conflict	Ridge sums	0.4554	0.1167	0/3	0.3084	0.001

Table 4: Final-date development results averaged over three event probabilities. Regret is the utility difference from the best evaluated option portfolio.

The relation GNN gives the strongest tariff ordering and reduces mean regret substantially relative to Deep Sets. No model recovers a held-out optimum in either event. The table is also a development result: an earlier execution exposed an input-scaling defect, and the correction was made after inspecting that behavior. A new, untouched date is needed for a prospective test.

8.3 Computation

American marks, exercise checks, and pathwise option sensitivities are computed once per contract. A portfolio then becomes a signed aggregation of the compiled leg paths, followed by its hedge and account calculation. This changes the compute comparison.

The benchmark contains 14 American options, 12,313 raw bounded combinations, 7,208 portfolios after account filters, 64 paths, and nine observations. Chain construction, scenario marks, and contract-leg compilation take 33.317 seconds and are shared by every method.

Candidates	Fixed-horizon aggregation	Daily-hedge aggregation
250	0.221 s	0.019 s
1,000	0.231 s	0.046 s
4,000	0.225 s	0.152 s

Table 5: Portfolio computation after the common contract work.

At 4,000 portfolios, direct daily-hedge aggregation takes 0.152 seconds. Deep Sets search with the 30% label budget takes 1.145 seconds after the shared phases; the relation GNN takes 13.763 seconds, including graph export, encoding, fitting, and scoring. End-to-end times are 33.470, 34.462, and 47.081 seconds. Deep Sets and the GNN obtain rank correlations of 0.9945 and 0.9920, respectively.

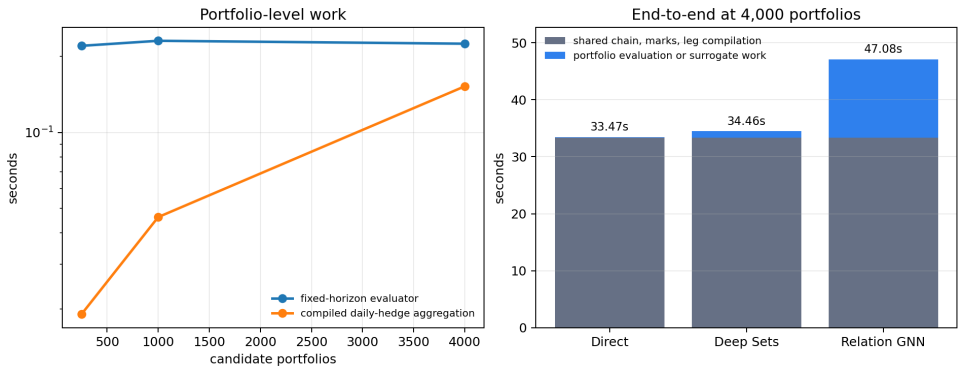


Figure 11: Compiler-aware runtime. Direct aggregation remains faster at 4,000 portfolios.

9 Conclusions

The empirical procedure begins with a point-in-time American option chain and ends with listed quantities, entry Greeks, bounded payoff, scenario P&L, lifecycle events, and a cash comparison. The directional studies show how a probability changes both the decision to trade and the preferred structure. The volatility studies answer a separate question: with median return fixed at zero, a target below entry IV favors bounded short gamma, while a target above entry IV favors long convexity after daily hedging and costs.

The learning results are more limited. Within one date, both neural representations fit the portfolio ranking well. Across dates, the ordering weakens and the best candidate is missed. Compiled direct evaluation is faster for the candidate counts measured here. More independent decision dates are therefore required before a learned shortlist can replace direct evaluation.

A later sequential study can admit conditional close, roll, exercise, hedge, inventory, and account actions. Those actions change later holdings, legal actions, and transaction costs; they are the setting in which a graph state and reinforcement learning may add value. That proposition requires its own empirical test.

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